Akshit Gupta

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Education

BITS Pilani, Hyderabad Campus CGPA: 8.12
B.E., Electrical and Electronics Engineering; Minor in Computing and Intelligence *2027*

Delhi Public School, Sushant Lok, Gurgaon 95.2% | 96.6%
Class XII | Class X (CBSE) *2023 | 2021*

Technical Skills

Languages: C++, Java, Python, C, SQL
Backend & Systems: Spring Boot, FastAPI, REST, microservices, Kafka, Redis, PostgreSQL, Oracle PL/SQL, Docker, AWS
Tooling & Testing: Git, GitHub Actions, GitLab CI/CD, CMake, Linux/macOS, JUnit, pytest, Playwright, ASan/UBSan
AI & Search: RAG, semantic/vector search, hybrid (dense + BM25) retrieval, reranking, LLM agents, tool calling, PyTorch
AI-Assisted Engineering: Claude Code for coding, debugging and test generation, with every change reviewed and tested
Coursework: Data Structures & Algorithms, Operating Systems, OOP, DBMS, Machine Learning, AI

Experience

Upsure.AI May 2026 – Jul 2026
AI/ML Engineering Intern

- Architected an event-driven **FNOL claims pipeline** (**Kafka**, Kogito rules on an **Apache AGE graph**, FastAPI state machine) routing cases to **ML models**, re-engaging users via SSE and queueing edge cases for time-bounded **human review**.
- Built the **telemetry SDK** and PostgreSQL **audit-logging** module for an internal observability platform, with **SHA-256 hash-chaining** for tamper-evident compliance logs and non-blocking instrumentation via daemon threads.
- Deployed a CPU-optimized **FastAPI** service (**ONNX Runtime** + RapidOCR) for damage detection and **multilingual OCR**: precision **0.56** → **0.83**, OCR latency ∼**14s** → ∼**4s**.
- Built a vehicle-capture pipeline on **Vision Transformers** (SigLIP + DINOv2 + ONNX orientation head), cutting latency **67% (82s to 27s)**, plus a **YOLO11s + DINOv2**-LogReg damage-detection ensemble (**ROC-AUC 0.893, FPR 0.009**).
- Integrated Gitleaks, Trivy and SonarQube into **CI** across multiple **GitLab** repositories, cutting manual testing time by **80%**; added a **code-fix agent** that **retrieves** relevant source files and tests LLM patches in isolated git worktrees.
- Built an **autonomous LLM QA agent** (**tool calling**, **structured outputs**) that writes Playwright, Postman and k6 test suites and **self-heals**: failing tests go back to the LLM, which patches them or **classifies** them as real bugs.

Coffeee.io May 2025 – Jul 2025
Software Development Intern

- Built an **AI-powered resume parser** in **Python** with **spaCy** and **NLTK NER** to extract data from **PDF/DOCX** files.

Projects

Crash-Safe Log-Structured Key-Value Store (C++20) | akshit9162.github.io/logkv

- Built an append-only, **CRC32**-protected **storage engine** in **C++20**: log-replay index rebuild and **torn-tail** crash recovery.
- Benchmarked three durability modes, measuring a **624**× throughput gap between no-fsync (**285K ops/s**) and full media flush (**456 ops/s**); batched **group commit** recovers **75**× of the gap with a bounded 128-record loss window.
- Verified crash consistency with **40 assertions**, bit-flip and torn-tail **fault injection** and a **SIGKILL harness** proving no acknowledged write is lost; **ASan/UBSan** clean in GitHub Actions CI.

Distributed Order Processing System | akshit9162.github.io/Distributed-Order-System

- Engineered **3 Java/Spring Boot microservices** (order, inventory, payment) with the **SAGA pattern** and compensating transactions, keeping independently owned **PostgreSQL** databases consistent.
- Guarded stock with **Redis distributed locks** and payments with **JPA optimistic locking** (**0 oversells** across 50 concurrent orders); **Kafka** dead-letter topics and idempotency keys; k6 **p95 137 ms** on **Terraform**-provisioned AWS.

Grounded RAG Platform | akshit9162.github.io/Grounded-RAG-Platform

- Built a **citation-grounded RAG** service (**FastAPI**, Claude) with **hybrid BM25 + dense retrieval** and **cross-encoder reranking**, so every answer cites its sources; **Dockerized** on AWS.
- Hit **1.0 context recall** and **0.947 answer relevancy** on **RAGAS**, with a **hallucination guardrail**.

Custom Linux Kernel: Fork-Bomb Detection and VIP Scheduler | C, Linux 6.14

- Modified the **Linux 6.14 kernel** in **C** to detect per-user **fork bombs** and extended the **CFS scheduler** with a priority-aware VIP class and starvation prevention, exposed via `/proc` and `sysctl`.
- **Benchmarked** scheduling fairness and overhead of the modified kernel under sustained load.

Achievements

- **Codeforces Expert** (max 1606) • LeetCode 1627, 400+ problems solved • **NTSE** Scholar, State Rank 155, Haryana.